

What a Frozen Detector Representation Resolves: Measurement Geometry, Training Support, and Consequence-Aware Diagnosis

Collaboration concept note (formerly *Consequence-aware failure diagnostics for learned particle-reconstruction representations*)

Dowling Wong

16 September 2026 (two-claim revision)

Abstract

A frozen reconstruction model deployed on a physics detector carries a contract: an assumed noise covariance and a training distribution. When either moves, a monitor must say which declared mechanism moved and whether the physics result is harmed. This note proposes a study that makes exactly two claims and tests both against a strong, committed generic monitor with identical alarm-time information. *Claim 1:* on declared, alarm-time-observable interventions, statistics of a model’s intermediate representations add acquisition-versus-valid-physics attribution information beyond a generic arm that already holds input quality, outputs, uncertainty, the final embedding, noise-only quality statistics and the same reference-distance transforms — measured as a paired macro-F1 difference on a hard, signature-matched evaluation set with capacity-matched and hook-drop controls, and with abstention on undeclared families evaluated empirically. *Claim 2:* a predeclared task-sensitive representation score, the length of a window’s deviation in the metric $J_y^\top W_y J_y$ pulled back from declared physics outputs, ranks scientific harm on held-out physical intervention cells better than the committed generic score — measured as a paired AUROC difference for $K \geq \kappa_m$ with the cell as the resampling unit. The measurement metric $\tilde{J}_g^\top \tilde{J}_g$ (the Gaussian Fisher information under the assumed covariance) and the task metric are kept apart; resolvability is not read as training support, which is estimated and validated separately; acquisition and support signatures are stated as conditional hypotheses that the development table already partly contradicts. The evidence ladder is deliberately short: a controlled waveform simulator in which the assumed and realized covariance are both known, run on a trained nonlinear subject, plus one independent transfer arm selected by readiness and licence gates; TIDMAD is an external check if its physical endpoint is defensible. A negative result — intermediate hooks do not justify their cost — is reportable if the positive controls, precision and baseline design pass. No result in this note is claimed; the protocol prototype, its tests and a non-citable development smoke exist, and no protocol is frozen.

1 Gap and contribution

The question. Cross-detector pre-training has shown that one sensor-level recipe learns useful representations on three physically different detectors, that those representations cut supervision by orders of magnitude, and that linear probes find physics in them [25]. That answers how general a *recipe*

can be. It does not say why a representation takes the form it does, or when it stops resolving the physics it was trained on. The companion theory [21] gives the determinants for a calibrated readout: the measurement fixes the metric Σ^{-1} , the architecture fixes the admissible class, and the training data fix the excited latent support; the wrong metric learns the wrong physical structure on planted data. This study asks the deployment half of the same question on frozen models: *which determinant moved, is the physics still resolved, and does it matter* — with the determinants supplying the predictions rather than a list of failure modes.

Existing shift monitors can detect changes in inputs, outputs, or embeddings, but a scientific reconstruction pipeline requires two additional distinctions: whether the change reflects corrupted acquisition or valid but under-supported physics, and whether the detected change materially degrades a physics result. We test whether intermediate representations improve this attribution beyond generic monitoring at the same false-alert budget, while allowing abstention on ambiguous or unseen failures. The attribution claim is conditional on the declared simulator and forward model: unknown physics and model misspecification are not identifiable from a response profile alone and are routed to abstention. No public dataset can decide these questions. None carries a known assumed-versus-realized covariance — simulation tools that *synthesise* correlated multichannel noise from a specified spectrum are open and recent ([22]; Wire-Cell’s per-band colorer matrices [23], on the measured coherent-noise model of [24]), but none reports the realized covariance of the ensemble it drew, which is what makes $\kappa(\hat{\Sigma}^{-1}\Sigma)$ a measured lever rather than a nominal setting — and none is event-paired across the factors a causal claim must vary: the released NuBench sets are independent per-geometry productions, and TIDMAD’s own authors disclaim the scientific meaning of its denoising score. The study therefore produces its own evidence in two designed datasets — a controlled-covariance waveform set (**ORACLE-Cov**) and a paired multi-geometry point-cloud set (**ORACLE-Paired**), the latter a citable artifact in its own right — and uses TIDMAD as the real-data check. The tiers are ordered: predictions fixed on the mechanism tier are pre-registered before the single confirmatory run on the realism tier.

In waveform reconstruction, structured acquisition noise induces a likelihood geometry through the inverse covariance: the correct residual weighting is Σ^{-1} , and a model evaluated under the wrong weighting can prefer reconstructions that are

worse in the metric the instrument implies. Architecture and training coverage further determine which physically relevant distinctions survive into the representation [21]. This study asks whether violations of these assumptions can be diagnosed after training, and whether a detected violation is scientifically consequential. The question has become concrete: cross-modality frozen backbones now exist — Panda V2 is pre-trained with one recipe on LArTPC, collider-TPC and water-Cherenkov data and adapted with 10^3 labels per detector [25] — and their authors list transfer across detectors and from simulation to real data as untested. Those are exactly the acquisition-shift and support-shift questions posed here, on a frozen public model. The artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a maintained software product.

Contribution, and what would refute it. Relative to the shift literature the study adds a setting and two paired tests, not a monitor. Failing Loudly [12] established that two-sample tests on a pre-trained classifier’s representation detect shift well and that a domain classifier characterizes it; shift malignancy there is classification accuracy. Zhang et al. [15] attribute a performance change to shifted mechanisms with labelled target data. Unlabelled performance estimation [13, 14] predicts a classifier’s aggregate target accuracy from confidences under stated assumptions on the shift. Conformal methods [4] give marginal coverage under exchangeability and no guarantee against arbitrary unknown shifts. What this study can establish beyond these is narrow and checkable: on a frozen physics reconstruction model under declared interventions with a known assumed covariance, (i) whether strictly internal representation statistics add attribution information beyond a generic arm holding the same alarm-time side information and the same distance transforms (Claim 1), and (ii) whether a task-sensitive representation score orders held-out intervention cells by a physics loss against evaluation-only truth better than a committed generic score, including harm that arrives without a strong alarm (Claim 2). Claim 1 is refuted by equivalence or detriment on the hard contrasts, or by a gain the ablations assign to input/output transformations, noise-only records, feature count or event identity; Claim 2 by a paired interval that includes zero without meeting the equivalence rule, by systematic low-alarm/high-harm failures, or by a gain that vanishes on held-out physical families. We do not claim a first or an exhaustive search; the components are established and cited as such.

Position relative to learned-geometry reconstruction. The subjects of this study are models that *learn* the geometry of the measurement: distance-weighted graph networks that place every sensor in a learned coordinate space [26], potential-based objectives that attach to each vertex a learned weight for whether it can represent its object [27], and cross-detector pre-training recipes [25]. None of these carries a statement of what its representation can resolve once the acquisition or the physics moves; recent benchmarks of physics foundation models report regime-dependent scores across distribution shifts without a mechanism [28]. The learned resolvability weight of object condensation is the closest relative of the resolvability diagnostic of Section 3

(the rank of M_{recon}), which computes a related quantity for a *frozen* model from its reconstruction Jacobian and the assumed covariance, with no training signal. Nothing here proposes a new reconstruction architecture; the contribution is the two paired tests and the designed data on which they can fail.

Identifiability limit. Diagonalizing a covariance yields decorrelated statistical modes, not labeled physical sources: $\Sigma = \Sigma_1 + \Sigma_2$ admits infinitely many positive-semidefinite splits, and low-rank factors are rotation-degenerate. All attribution here is over *declared intervention families* under a stated simulator. Observing a response profile does not identify the cause of an unknown real shift, and we make no such claim.

2 Failure contracts

Three declared labels, frozen before evaluation.

N—acquisition-contract violation. For this study the closed family is power-spectral-density (PSD) or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, and channel or module loss. The measurement violates a declared acquisition condition; no broader N claim is made.

S—training-support shift. Window or event composition moves toward physical strata that are sparse under the clean training distribution. The events themselves are clean and physically valid; only the support has moved.

E—evaluation-contract fault. Wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. This is a pipeline bug, not a distribution shift.

N versus S is the scientific core. E is decided by deterministic schema and range invariants, so it is reported as a gate result, not as a class in a classification score. The informative and difficult separation is N from S. N and S may both move the observed input distribution, so generic input-shift detection alone cannot distinguish them. Evidence for N comes from acquisition-quality indicators, paired corruption interventions, or known violations of the measurement model. S is defined by physically valid, uncorrupted events that occupy low-density regions of the clean training support. The scientific problem is therefore not whether the input changed, but whether the change reflects corrupted measurement or valid but under-supported physics. Section 3 states why the two leave different geometric signatures in a frozen representation, which is what makes N versus S a prediction rather than a label. That distinction is what matters operationally: discarding S rejects physically valid events, whereas rejecting N removes measurements that violate the declared acquisition contract.

3 Two metrics, and what a frozen representation can resolve

Write the declared forward model as $x = g(r) + n$, $n \sim \mathcal{N}(0, \Sigma)$, where $x \in \mathbb{R}^{C^N}$ is the sampled measurement, $r \in \mathbb{R}^d$ the representation being monitored (for the analytic subject, the pooled latent z ; for a network, any hooked stage), g the reconstruction back into measurement space and $y = h(r) \in \mathbb{R}^t$ the map to the declared physics outputs (amplitude, arrival time, shape). Two pullback metrics live on r , and they are different objects.

Measurement metric. Under the assumed covariance $\hat{\Sigma}$ the Fisher information of r is $I(r) = J_g^\top \hat{\Sigma}^{-1} J_g$ with $J_g = \partial g / \partial r$ — the standard Gaussian-model result [19]; in whitened coordinates $\tilde{J}_g = \hat{\Sigma}^{-1/2} J_g$ it is $M_{\text{recon}} = \tilde{J}_g^\top \tilde{J}_g$, unit-free (whitened residual per sample), local to the reference point, and dependent on the *assumed* covariance — the operational one — never the realized Σ . Its eigen-directions above a rank threshold are the directions the measurement *resolves*; those below are weakly resolved and, where $\text{rank } M_{\text{recon}} < d$, not locally identifiable at all. That is a local identifiability statement. It is not a statement about which directions the training distribution excited, and this paper no longer reads it as one.

Task metric. With $J_y = \partial y / \partial r$ and a declared positive semi-definite weight W_y in physics-output units (a diagonal of inverse squared resolution requirements, or a one-hot selecting the consequence target), $M_{\text{task}} = J_y^\top W_y J_y$ measures how far a representation displacement moves the declared outputs. Σ does not enter: J_y maps to outputs, not to measurements, so an expression of the form $J_y^\top \Sigma^{-1} J_y$ is dimensionally invalid and is not used. The task-sensitive score of Claim 2 is the length $\|r - \bar{r}\|_{M_{\text{task}}}$ of a window’s deviation from the reference mean, standardized on clean windows. Neither metric measures realized scientific harm; Claim 2 tests that connection empirically.

Training support is estimated, not inferred from rank. Whether a window lies where the training distribution placed mass is estimated by a separate score on the reference representation — a shrinkage Mahalanobis distance combined with a k -nearest-neighbour distance, each standardized by its clean reference quantile — validated before use on constructed in-support and out-of-support controls, and used only when that control passes a declared floor. Valid rare physics can therefore be *in-support-but-rare* or *outside estimated support*; the development table already contains both kinds (Section 5).

Conditional signatures, not a dichotomy. The earlier reading that acquisition shift stays in the excited span while support shift lives in its complement is contradicted by the development table: gain drift, channel loss and timing jitter — acquisition changes by contract — produce representation shifts with out-of-span fractions near one, and in-span rare physics moves the representation a great deal while leaving every noise-only statistic at its reference. The mechanism section therefore states conditional signatures, each a hypothesis the attribution arms test: (i) a covariance-only acquisition change ($\hat{\Sigma} \neq \Sigma$, g unchanged) changes noise-only residual statistics — whitened variance, residual spectrum, residual channel correlation on random-trigger records — without a consistent representation mean shift; (ii) a structural acquisition change (gain, channel loss, jitter) can produce both a representation shift and a residual shift; (iii) valid rare physics leaves the noise-only statistics at their reference and moves the representation either inside the resolved span (head-limited) or outside estimated support; (iv) origin and harm are separate axes. The signatures do not prove causal origin outside the declared intervention design; the classifier of Claim 1 tests whether their collection, computed from intermediate hooks, adds information beyond the generic

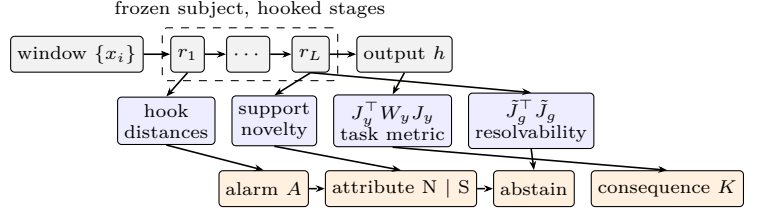


Figure 1: Monitor stack on a frozen subject. Hooked stages feed the intermediate-hook statistics (reference distances of pooled stage means and second moments, reference principal coordinates); the output map feeds the task metric M_{task} and, through the reconstruction, the measurement metric M_{recon} . Alarm, N/S attribution, abstention and the harm measure K are separate endpoints; abstention is a support novelty with a clean-window conformal threshold.

arm that already sees (i)–(iii) at the input, output and final embedding.

Invariance, stated exactly. Every reported statistic is labelled with the group under which it is invariant. Reference Mahalanobis distances with shrinkage are invariant to orthogonal maps and isotropic rescaling of a hook, not to shear; energy splits in reference projectors are invariant to orthogonal maps and isotropic scale; the task length is a pullback and is invariant to any invertible reparameterization under which J_y transforms covariantly. Singular subspaces of stage Jacobians are invariant to orthogonal basis changes only. Each statement is a unit test with an orthogonal, a scaling and a shear control; where a dependence exists it is reported as a limitation, not hidden behind “gauge-safe”.

Companion-paper dependency, bounded. The companion preprint [21] develops the covariance-weighted linear classes and the excited-support argument. Nothing central here rests on it: the measurement metric is the textbook Fisher information [19], the task metric is defined above, and the support and signature statements are empirical hypotheses tested by this study. Where the companion result is cited it is cited as a preprint, with its proposition numbering left unstated until the posted version is fixed.

4 Two claims

The study makes exactly two primary scientific claims, each with one predeclared paired estimand. Everything else in this note — detection power, monitoring cost, physical-variable probes, the designed perturbation controls, resolvability and patching analyses — is a supporting analysis or a positive control, reported where useful and never numbered as a claim.

Claim 1 — incremental attribution value. On declared, alarm-time-observable interventions, do intermediate representations improve acquisition-versus-valid-physics attribution beyond a strong generic monitor that already receives every operational input-quality, output, uncertainty, final-embedding and noise-only feature *and the same reference-distance transforms*? The estimand is the paired difference on a frozen *hard* evaluation set of N and S windows with overlapping generic signatures,

$$\Delta F1_{\text{attr}} = F1(\text{full_intermediate}) - F1(\text{generic_rich}),$$

macro-F1 over $\{N, S\}$, where **generic_rich** is the committed generic arm and **full_intermediate** adds only information

genuinely absent from it: statistics of strictly internal hooks, with no whitened-input, final-embedding, pre-output or output duplicate. Controls: a generic arm with matched feature count (a quadratic expansion of its own scalars), hook-drop arms, and the internal hooks alone. Provisional reading, unfrozen until justified against achievable precision: benefit needs a 95% interval above zero and a point estimate of at least 0.10; equivalence needs the whole interval inside ± 0.05 ; a set below a declared minimum size yields no reading. Required supporting results: inclusive and hard-matched results with retained fractions; confusion matrices; family and held-out-severity results; noise-only and capacity ablations; operational versus privileged variants; selective risk and retained coverage under abstention; and a joint operational table over clean, N, S, mixture and unknown windows, so that a good conditional N/S classifier is not mistaken for a deployed decision rule. *Refutation*: equivalence or detriment on the hard contrasts, or a gain attributable to extra input/output transformations, noise-only records, feature count, event identity or unavailable side information.

Claim 2 — incremental harm ranking. Does a predeclared task-sensitive representation score rank scientific harm better than the committed generic score on held-out, physically generated intervention cells? With K an independent physical endpoint (amplitude error against evaluation-only truth on the waveform tier; angular error on a neutrino-telescope subject), harm $K \geq \kappa_m$, and both scores fixed without evaluation labels,

$$\Delta \text{AUROC}_{\text{harm}} = A_{K \geq \kappa_m}(\text{task}) - A_{K \geq \kappa_m}(\text{generic}),$$

with $A_{K \geq \kappa_m}(\cdot)$ the AUROC of a score for harm over all held-out cells under a frozen cell weighting. The generic score is committed in advance as the maximum of clean-calibrated reference distances of the whitened input, the outputs and the final embedding; the task-sensitive score is the clean-calibrated task length of Section 3. Selecting the better of several generic scores after evaluation is prohibited. The interval resamples the unit the claim generalizes over — the intervention cell — with event groups nested inside and model seeds as declared; with too few cells the interval is reported as descriptive and no confirmatory inference is made. Required supporting results: both absolute AUROCs, AUPRC with prevalence, the four alarm-harm quadrants with counts at a cell-level threshold calibrated on clean windows, the missed-harm rate at the alert budget, false rejection of benign valid-rare cells, family and severity breakdowns; the conditional strong-alarm triage is secondary and reports how many cells the conditioning dropped. κ_m must be declared from a scientific requirement before evaluation; it is *pending* on every arm, development runs use a provisional 10% amplitude degradation and say so, and the confirmatory runner refuses while any κ_m is pending or any freeze, environment, data or model hash is missing. *Refutation*: a paired interval including zero without meeting a predeclared equivalence rule, systematic low-alarm/high-harm failures, or a gain that disappears on held-out physical families.

Table 1: Claim 2: alarm-harm quadrants and endpoints (cell level).

	$K < \kappa_m$	$K \geq \kappa_m$
Low alarm	quiet/benign	missed harm
Strong alarm	harmless alarm	detected harm

Primary: paired $\Delta \text{AUROC}_{\text{harm}}$ over all held-out cells, cell-outer resampling. **Supporting:** absolute AUROCs; AUPRC with prevalence; missed-harm rate at the budget; benign valid-rare cell rejection (and the window-level estimator); breakdowns; conditional strong-alarm triage with the number of cells dropped. All four quadrants are reported with counts.

Supporting analyses. *Detection*: power at a 1% false-alert rate on non-overlapping clean windows and delay in windows, with a binomial interval on the realized rate — a hundred clean windows resolve 1% in one-percent steps only, so calibration counts are an output of a development precision study. *Cost*: latency and storage of sampling and sketching against full capture, measured on a named machine. *Probes*: ridge probes from hooked representations to the tier’s physical variables, reported descriptively; probe accuracy is never read as a physical axis. *Designed controls*: output-null, output-aligned, random and task-specific aligned/null families, constructed from the frozen head and therefore positive controls for any monitor built from that head; exact only for the tied linear subject, refused for nonlinear subjects until a constrained local inverse with an input-validity check exists; norm matched in the alarm metric being challenged; realized physical loss reported per output, with the local-linear error. *Resolvability and patching*: the rank of M_{recon} and activation patching are mechanistic diagnostics; uniform recovery at every stage, which the linear subject shows, is not localization.

Abstention. Abstention is a feature-novelty rule: the support score of Section 3 with a threshold set by split-conformal calibration on clean calibration windows [4]. Under exchangeability of those windows with deployment *clean* windows the retained fraction of clean windows is at least $1 - \alpha$ in expectation; the rule guarantees nothing about arbitrary unknown families, so unknown-family detection is measured empirically (AUROC), together with the risk-coverage curve and its area, retained coverage at the threshold, and retained-known macro-F1 with counts [5]. Unknown families appear only in the evaluation partition. Retained coverage is a reporting point, not a threshold tuned on confirmatory labels.

Identifiability and origin. All attribution is over declared intervention families under a stated simulator; observing a response profile does not identify the cause of an unknown real shift, and such windows are routed to abstention. Origin (N, S, geometry, mixture, unknown) and harm (benign, harmful) are scored as separate axes: a clean rare event can be harmful and a mild corruption benign.

5 Testbeds and monitoring design

The three tiers are three physically different measurement systems — cryogenic-style waveforms with a calibrated covariance, neutrino-telescope point clouds with no covariance at all, and a SQUID time series with recorded truth — under one principle. Generality is claimed across measurement regimes, not across architectures.

Tier 2 — neutrino-telescope point clouds (realism: ORACLE-Paired). One primary public configuration with a released checkpoint. The intermediate arm hooks all four DynEdge convolution blocks plus the post-processing

and readout stages, not only the 128-dimensional pre-head event representation. The data are our own production: the released NuBench datasets are independent per-geometry productions and share no latent events, so the paired detector-geometry contrast is constructed by re-simulation — Prometheus [18] re-uses one seeded injection across detector geometries (here ORCA and ARCA, both water, so layout is not confounded with medium and the photon stage is exactly reproducible), with the NuBench detector-response emulation reimplemented in-project, which is precisely where the acquisition knobs live. Per geometry, from one injection: module dropout, hit thinning, time jitter, gain drift and noise-rate cells instantiate N ; S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training data only — with the caveat that module loss *lowers* observed multiplicity, so every N cell is matched by clean events at the same observed multiplicity, charge, and time spread; otherwise attribution reduces to recognizing the corruption operator’s fingerprint. Undeclared families (double-bang ν_τ , through-going muons, pile-up, correlated module noise) are held entirely outside the declared contract for the abstention endpoint, and the injection identifier is the pairing key across geometries. One secondary architecture serves as a consistency check. Until the paired set exists, the released Hexagon water/ice pair supports only a distribution-level medium comparison, and is labelled as such. No assumed covariance exists in this tier — deliberately: it tests, out of sample, the monitor-failure predictions pre-registered on Tier 1, and it is where the one alarm computable *without* knowing Σ (the task length in M_{task} , which needs only J_y) earns its deployability claim.

Tier 1 — controlled waveforms (mechanism: ORACLE-Cov). The synthetic arm uses a noise-aware linear control and a generator that constructs PSD/cross-spectral density models, samples seeded frequency-domain coefficients with real-signal symmetry, and adds declared artifacts; the generator reports both the implied and the realized covariance of every ensemble. The synthesis itself is not claimed as novel: `pytessim` draws from a cross-spectral density by per-frequency Cholesky factorisation [22], and Wire-Cell colours per frequency band with user-supplied matrices A satisfying $AA^\top \approx \Sigma$ [23]. Reporting the *realized* covariance of the drawn ensemble, so that $\hat{\Sigma} \neq \Sigma$ is measured rather than nominal, is what neither does and what this tier requires. The inverse-PSD-weighted training objective makes the encoder’s assumed covariance $\hat{\Sigma}$ a literal property of the loss, so $\hat{\Sigma} \neq \Sigma$ is a controlled experimental lever, swept through the condition number $\kappa(\hat{\Sigma}^{-1}\Sigma)$. Every theoretical claim is decided here, and the power analysis that sizes the other tiers runs here, because only here is the null exactly simulable. Development runs on an analytic linear subject (September 2026; not results of the study) shape the arms above: noise-only statistics alone separate covariance-type acquisition changes from physics changes, covariance mismatches of condition number 5–12 produce large alarms with little consequence on that subject, in-span rare physics raises the consequence while leaving every noise-only statistic at its reference, structural acquisition changes move the representation with out-of-span fractions near one, and activation

patching is flat — so stage localization is a question for the nonlinear subject, and any attribution gain of an intermediate arm must be shown to exceed what the generic arm with noise-only records and the same distance transforms already gives. A first development smoke of the two-claim protocol on the linear subject (16 September 2026; non-citable, toy size) found no reading on its 20-window hard set, a small inclusive gain that a capacity-matched generic arm largely absorbed, and a Claim-2 difference whose cell-outer interval included zero.

Tier 3 — TIDMAD (real data). The public TIDMAD [29] arm uses a $\sim 0.6\text{M}$ -parameter two-stage domain transformer, comparing otherwise identical MSE and inverse-PSD-weighted objectives — two assumed covariances on identical real electronics noise, the naturally occurring real-data echo of the Tier-1 contrast. The release’s denoising score is a development consequence only; the confirmatory consequence is the relative exclusion-limit comparison. This compact transformer is a diagnostic subject, not a foundation model; a larger pretrained model is only a later gated comparison.

Monitoring comparison. Every arm consumes only features available at alarm time, built through a typed input object that carries the observed window, its random-trigger records and the geometry — and nothing else — so that truth, the paired replay, the realized covariance and the intervention label are not arguments of any alarm-time builder; every feature carries its source tags and an arm is scored only from a locked, tag-checked batch. (A builder that lies about a tag is not caught; code review remains part of the contract.) One regularized logistic classifier, one tuning grid and one set of partitions serve every arm. Partitions are five, split by event group so that every geometry, replay and corruption variant of an event stays together: reference fitting (clean subject, projectors, nulls — never supervised examples), attribution training, development (tuning), clean calibration (false-alert budget, conformal threshold) and evaluation, scored once. Held-out families, severities and seeds are declared and scored on evaluation groups only; undeclared families enter no fit, tuning or calibration. The primary arms are `generic_rich` (input quality, outputs, an uncertainty proxy, the final embedding, operational noise-only statistics, and reference distances of the whitened input, the outputs, the pre-output, the energy split of the final embedding, the out-of-span fraction and the support novelty), `intermediate_only` (reference distances of the pooled mean and second moment of each internal hook and its reference principal coordinates) and `full_intermediate`; controls are a capacity-matched generic arm and hook-drop arms. Noise-only statistics are operational only where the acquisition supplies random-trigger records (the controlled simulator, the HeST arm and TIDMAD’s no-injection files do; hit-level Prometheus output does not, and such an arm is reported as privileged). Reference distances use Ledoit–Wolf shrinkage [20] and, where a hook’s dimension exceeds a fifth of the reference count, a reference-fitted PCA; each is mapped to a common clean-null scale before any combination, and their stability across reference sample size and rescaling is tested — at the reference sizes of a development smoke they

are not stable, which the report states. Hard contrasts are built by 1:1 matching of S to N windows inside a declared caliper on standardized generic signatures, with retained fractions and overlap failures reported; matching is a declared conditional design, not a causal identification. Eligible base-lines — corrected univariate input tests, RBF-MMD [6], a classifier two-sample test [7], embedding mean/covariance distance, output tests, uncertainty/entropy — receive identical windows and false-alert budgets. Clean miscalibration is a model result, not automatically a software bug [8]. Intervals resample event groups for window-level estimands and intervention cells for cell-level ones, with model seeds as a declared level; null results require a passed positive control, adequate precision and a predeclared equivalence margin.

Reporting standard. Every learned arm and baseline is reported with its parameter count, training cost, and inference time and memory per event, and arms compared against each other are matched in parameter count, not in tuning effort. Every endpoint is reported inclusively and on the stratum where the methods are predicted to differ (highest severity, lowest SNR, boundary support), under the same metric names on every tier. Every threshold — alarm, attribution, abstention — is reported as a curve at the pre-registered false-alert budget (risk-coverage for abstention), never as a single operating point. Each learned ladder contains an extrapolation arm trained inside and evaluated outside the declared severity and multiplicity range, labelled as such. The hyperparameters of every method appear in one table with the values used.

Table 2: Experiment matrix. The two claims run on the trained Tier-1 subject and one transfer arm; the rest are supporting analyses.

ID	Test	Endpoints
E0	Clean replay; deliberate evaluation-contract faults (EF)	Reference discrepancy; gate sensitivity/specificity (gate, not a claim)
A1	Claim 1: hard-matched N/S attribution, primary arms and controls, abstention, joint decision table	$\Delta F1_{\text{attr}}$ with interval; ablations; unknown AUROC; risk-coverage; joint table
A2	Claim 2: harm ranking on held-out cells	$\Delta \text{AUROC}_{\text{harm}}$ with cell-outer interval; quadrants; missed harm; valid-rare rejection
S1	Detection power and delay at 1% FAR	power, delay, realized FAR with binomial interval (supporting)
S2	Designed and task-specific controls	realized per-output loss at matched norm; local-linear error (positive control)
S3	Probes, resolvability, patching	probe R^2 ; rank of M_{recon} ; patched- K recovery (mechanistic, descriptive)
S4	Capture vs sample/sketch	p50/p95 latency; bytes/event; loss (supporting)
T1	Transfer: the two claims repeated once on one held-out physical arm	same estimands, predeclared; failures reported as failures

6 Evidence ladder and roadmap

Minimum viable paper. (1) The two claims on the controlled waveform tier with a *trained nonlinear* subject; (2) the two claims repeated once on one independent transfer arm chosen by readiness, licence and physical- K gates (HeST $\rightarrow$ our readout is built; LUCiD is licence-gated; Prometheus is the fallback with a frozen public model and angular error as K); (3) the paired comparisons with their failures and limitations. TIDMAD is an external check if its relative exclusion-limit endpoint survives scrutiny.

Phase A — theory and schema gate (this revision). Two metrics separated; resolvability renamed; support estimated; conditional signatures; invariance controls; protocol layer hardened; development smoke. No trained campaign starts before adversarial review of this phase.

Phase B — precision and feasibility (development only). Calibration size for the false-alert budget; outer-unit counts for Claim 2; model-seed variability; matched-contrast overlap. Unachievable precision changes the claim or the design before any freeze.

Phase C — scientific evidence. The trained Tier-1 subject in development mode; if controls and hard contrasts pass, exactly one transfer arm; freeze only after thresholds, hold-outs, cell weights, scores, margins, counts and dependency hashes are agreed.

Future work, not this paper. Additional geometries, the linear estimator classes as probes of physical axes, sampling-cost optimization, Panda/SPINE transfer, and stage localization, unless one of them directly resolves a claim.

Relation to chained-model uncertainty propagation. Douglas et al. [3] propagate input uncertainty through a chained neutrino-LAr reconstruction and compare uncertainty-aware and blinded models under synthetic noise. Here the model is frozen: the question is whether its representation attributes a declared failure, abstains outside scope, and links an alarm to consequence. Propagated uncertainty is therefore a candidate covariate, not the endpoint.

Relation to shift monitoring and localization. The components of this study are individually established, and we claim the physics instantiation and the evaluation protocol, not the components. Shift-type identification from a frozen encoder exists in medical imaging [9], building on the acquisition-versus-population taxonomy of [10]; harmful-versus-benign shift detection is a named research line [11, 12], and performance changes have been attributed to specific shift mechanisms [15]. That shift type determines *which layers* to adapt is the headline of surgical fine-tuning [16], and localization-then-edit validation is known to be a weak inference [17], which is why any later repair experiment here is preceded by an activation-patching causal check rather than resting on adaptation gains. Unlabelled performance estimation [13, 14] predicts a classifier’s aggregate accuracy on a shifted distribution from confidences, under assumptions on the shift that its authors state cannot be verified from unlabelled data alone. What none of these supply, and what this study tests, is (i) a consequence variable that is a physics observable under a known measurement covariance rather than classification accuracy, (ii) the paired all-cell alarm-harm comparison of Table 1, including harm that arrives without a strong alarm, with designed dissociations as positive controls, and (iii) an attribution comparison whose generic arm already holds the acquisition’s own noise-only statistics and the same distance transforms as the intermediate arm. We are not aware of a monitoring result for learned neutrino-telescope reconstruction and do not claim

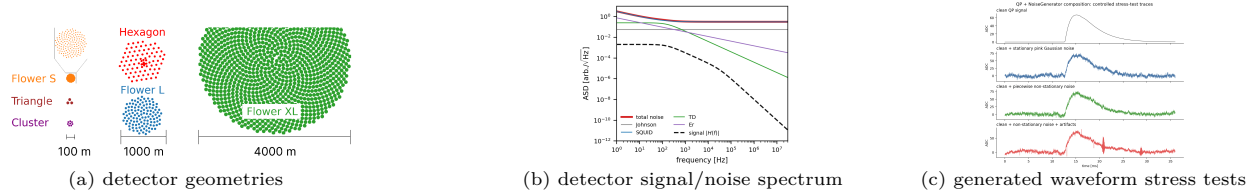


Figure 2: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The analytic $\text{Al}_2\text{O}_3/\text{Al}$ athermal reference budget — component and total noise ASDs with the two-pole signal response — regenerated from the parameterised model in `noise_module.reference_budget`. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

the absence of one.

7 Collaboration

The agreed sequence is $B \rightarrow A \rightarrow A \oplus B$, while the waveform/TIDMAD arm runs in parallel and supplies the first code-complete contrast. Work is consolidated in the shared ORACLE repository. A proposed fortnightly meeting reviews one frozen protocol or result; initial ownership is waveform infrastructure and generic monitoring on Dowling’s side, and geometry/SPINE physics choices and validation on Junjie’s side, with thresholds and claim changes approved jointly. The first meeting should freeze one geometry intervention, pairing variables, one representation endpoint, and one downstream physics endpoint.

8 Feasibility gate

Dataset access, checkpoint loading, released-prediction rescoring, intermediate-hook extraction, and a monotone module-dropout positive control have been validated ($15.28^\circ \rightarrow 25.12^\circ$ median angular error and 10-NN retention $1.000 \rightarrow 0.393$ across 0–50% dropout), with the caveat that the pilot sample was enriched and non-randomly drawn; the audited scripts re-draw it uniformly with nested severities and paired intervals. Exact re-inference parity remains unresolved: the current pipeline differs from the released directions by 0.944° median angular error. Two identified confounds are resolved first: the released-metric rescoring had grouped by the track/cascade topology flag while Table 6 categories are interaction classes, and the pilot substitutes five preprocessing methods of the released detector, which is itself a candidate cause and is now disclosed with the outputs. The decisive next test is cheap: identical re-inference on CPU and GPU separates a k -NN backend cause ($\text{CPU} \neq \text{GPU}$) from preprocessing, alignment, or version skew ($\text{CPU} = \text{GPU} \neq \text{release}$). Exact identity is a software control, separate from the scientific geometry gate. On a fixed development subset we will test whether the residual offset is stable across severity and event strata. Stable offsets permit a predeclared backend tolerance for the paired contrast; severity-dependent drift is a confound and keeps the gate closed.

References

- [1] R. F. Ørsøe et al. “NuBench: An Open Benchmark for Deep Learning-Based Event Reconstruction in Neutrino Telescopes.” *JINST* 21 (2026). [arXiv:2511.13111](#).
- [2] D. H. Koh et al. (DeepLearnPhysics). “Scalable, Proposal-free Instance Segmentation Network for 3D Pixel Clustering and Particle Trajectory Reconstruction in LArTPCs.” [arXiv:2007.03083](#) (2020).
- [3] D. Douglas, A. Mishra, F. Petersen, D. Ratner, K. Terao. “Uncertainty Propagation within Chained Models for Machine Learning Reconstruction of Neutrino-LAr Interactions.” [arXiv:2411.09864v3](#), preprint (2025).
- [4] A. N. Angelopoulos and S. Bates. “A Gentle Introduction to Conformal Prediction.” [arXiv:2107.07511](#) (2021).
- [5] Y. Geifman and R. El-Yaniv. “Selective Classification for Deep Neural Networks.” *NeurIPS* (2017).
- [6] A. Gretton et al. “A Kernel Two-Sample Test.” *JMLR* 13 (2012), 723–773.
- [7] D. Lopez-Paz and M. Oquab. “Revisiting Classifier Two-Sample Tests.” *ICLR* (2017).
- [8] C. Guo et al. “On Calibration of Modern Neural Networks.” *ICML* (2017).
- [9] M. Roschewitz, R. Mehta, C. Jones, B. Glocker. “Automatic Dataset Shift Identification to Support Safe Deployment of Medical Imaging AI.” *MICCAI* (2025). [arXiv:2411.07940](#).
- [10] D. C. Castro, I. Walker, B. Glocker. “Causality Matters in Medical Imaging.” *Nature Communications* 11, 3673 (2020).
- [11] A. Podkopaev, A. Ramdas. “Tracking the Risk of a Deployed Model and Detecting Harmful Distribution Shifts.” *ICLR* (2022). [arXiv:2110.06177](#).
- [12] S. Rabanser, S. Günnemann, Z. Lipton. “Failing Loudly: An Empirical Study of Methods for Detecting Dataset Shift.” *NeurIPS* (2019).
- [13] D. Guillory, V. Shankar, S. Ebrahimi, T. Darrell, L. Schmidt. “Predicting with Confidence on Unseen Distributions.” *ICCV* (2021). [arXiv:2107.03315](#).
- [14] S. Garg, S. Balakrishnan, Z. C. Lipton, B. Neyshabur, H. Sedghi. “Leveraging Unlabeled Data to Predict Out-of-Distribution Performance.” *ICLR* (2022). [arXiv:2201.04234](#).
- [15] H. Zhang, H. Singh, M. Ghassemi, S. Joshi. “‘Why Did the Model Fail?’: Attributing Model Performance Changes to Distribution Shifts.” *ICML* (2023). [arXiv:2210.10769](#).
- [16] Y. Lee, A. S. Chen, F. Tajwar, A. Kumar, H. Yao, P. Liang, C. Finn. “Surgical Fine-Tuning Improves Adaptation to Distribution Shifts.” *ICLR* (2023). [arXiv:2210.11466](#).
- [17] P. Hase, M. Bansal, B. Kim, A. Ghandeharioun. “Does Localization Inform Editing? Surprising Differences in Causality-Based Localization vs. Knowledge Editing in Language Models.” *NeurIPS* (2023). [arXiv:2301.04213](#).
- [18] J. Lazar, S. Meighen-Berger, C. Haack, D. Kim, S. Giner, C. A. Argüelles. “Prometheus: An Open-Source Neutrino Telescope Simulation.” *Comput. Phys. Commun.* (2024). [arXiv:2304.14526](#).
- [19] S. M. Kay. *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall (1993), ch. 3 (the Cramér–Rao bound and Fisher information for the Gaussian model).
- [20] O. Ledoit and M. Wolf. “A well-conditioned estimator for large-dimensional covariance matrices.” *J. Multivariate Anal.* 88 (2004), 365–411.
- [21] D. Wong. “Noise Covariance Defines Reconstruction Geometry: Optimal Filtering, Weighted Subspaces, and Tied Linear Autoencoders.” Preprint (2026), unpublished at the time of writing; cited for the covariance-weighted linear classes only, no central result of this note depends on it.
- [22] TESSERACT Collaboration. “pytessim: simulation of TES datastreams including noise, LEE, and signals.” MIT-licensed software, first public release 2026-05-18. [github.com/spice-herald/pytessim](#). Multichannel noise sampled from a PSD or cross-spectral density by per-frequency Cholesky factorisation (`core/noise/Noise_Factory.py`).
- [23] Wire-Cell Team. “Wire-Cell Toolkit.” LGPL-licensed software. [github.com/WireCell/wire-cell-toolkit](#). `gen/src/CorrelatedAddNoise.cxx` (added 2026-02-04) colours per frequency band with user-supplied matrices A satisfying $AA^T \approx$ the correlation matrix; `CoherentAddNoise` with `GroupNoiseModel` gives shared-private block structure.
- [24] MicroBooNE Collaboration. “Noise Characterization and Filtering in the MicroBooNE Liquid Argon TPC.” *JINST* 12, P08003 (2017). [arXiv:1705.07341](#).
- [25] S. Young, C. Jesús-Valls, K. Terao. “Panda Diplomacy: Foundation Model Pre-training across Particle Imaging Detectors for High Energy and Nuclear Physics.” [arXiv:2609.00611](#) (2026).

- [26] S. R. Qasim, J. Kieseler, Y. Iiyama, M. Pierini. “Learning representations of irregular particle-detector geometry with distance-weighted graph networks.” *Eur. Phys. J. C* 79, 608 (2019). [arXiv:1902.07987](#).
- [27] J. Kieseler. “Object condensation: one-stage grid-free multi-object reconstruction in physics detectors, graph and image data.” *Eur. Phys. J. C* 80, 886 (2020). [arXiv:2002.03605](#).
- [28] M. Chu, Y. Liu, A. Biswas, H.-W. Shen. “Do Physics Foundation Models Learn Generalizable Physics? A Bias-Aware Benchmark Across Physical Regimes and Distribution Shifts.” [arXiv:2605.29283](#) (2026).
- [29] J. Fry, X. Fu, Z. Fu, K. Pappas, L. Winslow, A. Li. “TIDMAD: Time-Series Dataset for Discovering Dark Matter with AI Denoising.” *NeurIPS Datasets & Benchmarks* (2025). [arXiv:2406.04378](#).